

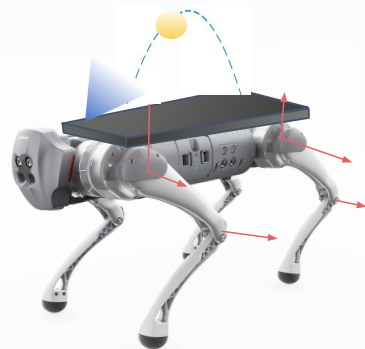
Hierarchical Planning for Quadrupedal Juggling

Combining RL Policies and Classical Control on a Unitree Go1

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Hybrid Robotics Lab · UC Berkeley

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Outline

1. Problem Statement
2. Motivation
3. Background / Prior Work
4. Methodology
5. Results

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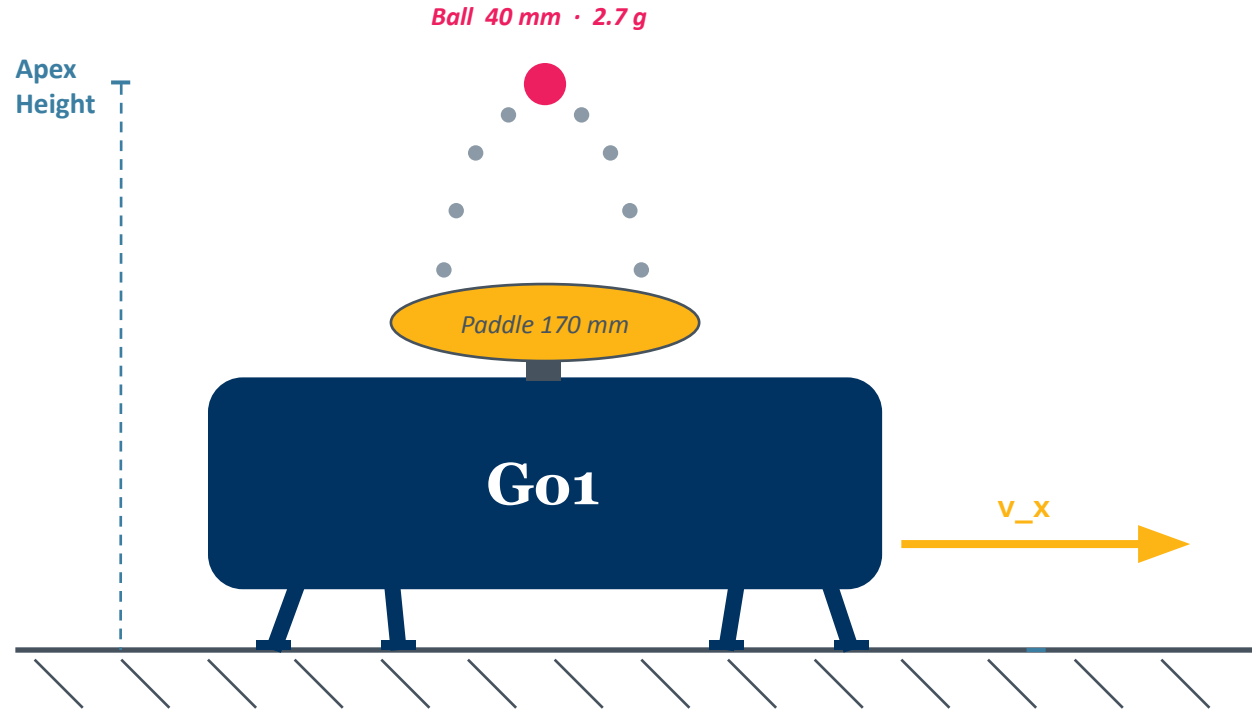
- 1. Problem Statement** ← *next*
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Problem Statement

Task: *A Unitree Go1 must bounce a 40 mm, ~2.7 g ping-pong ball steadily on a 170 mm disc paddle affixed to its back.*

Input: Velocity, yaw, and apex height

Simplified System Diagram



Problem Statement

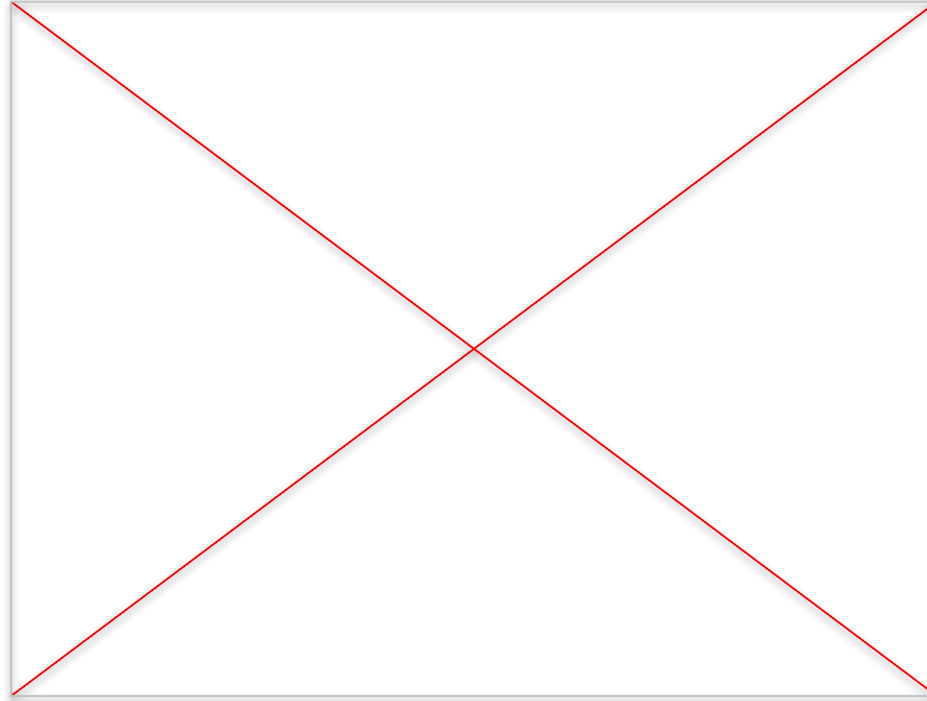
Motivation

Background / Prior Work

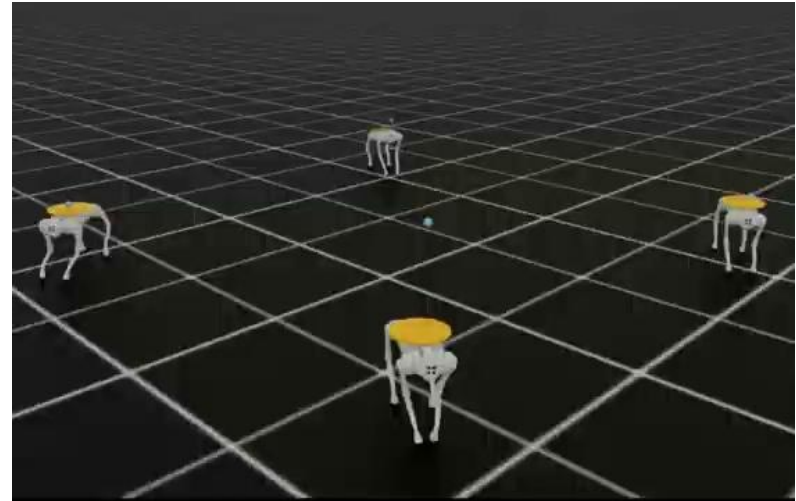
Methodology

Results

Results Preview - Torso Tracking



Results Preview - Stationary Juggling



Results Preview - Moving while Juggling...



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Motivation

Why sports related tasks?

Progress transfers to arbitrary challenging, dynamical tasks: healthcare, manufacturing, etc.

Our Thesis

A simplified, physics-based control layer can be productively combined with a pre-trained RL locomotion policy to produce a capable and interpretable hybrid system.

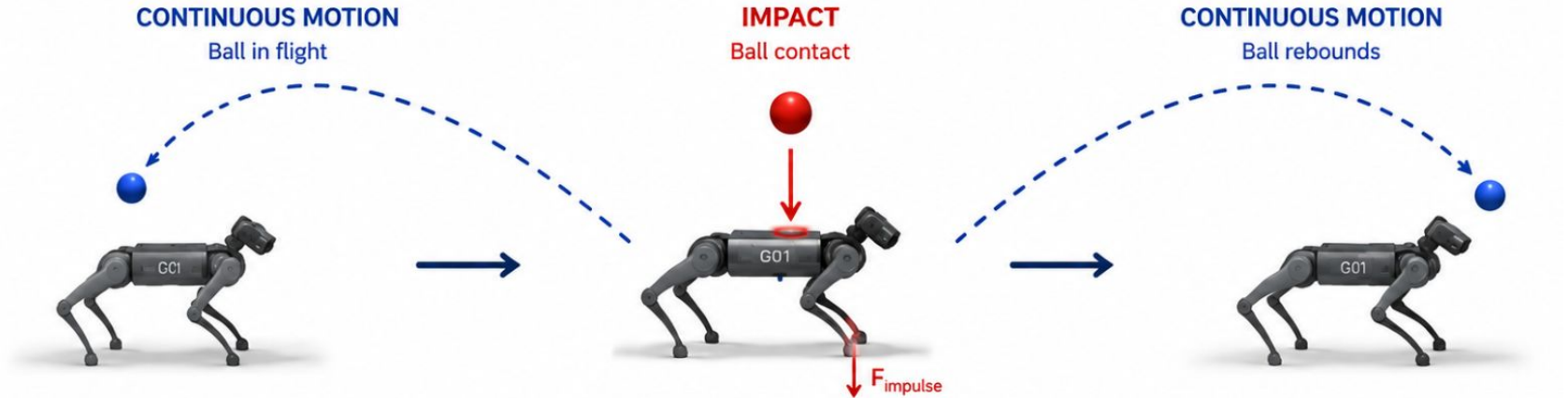
Why Juggling?

Fast closed-loop
dynamics

Hybrid contact-rich
behavior

Sensitive to
perception and
estimation errors

Natural benchmark
for robot
coordination



Problem Statement

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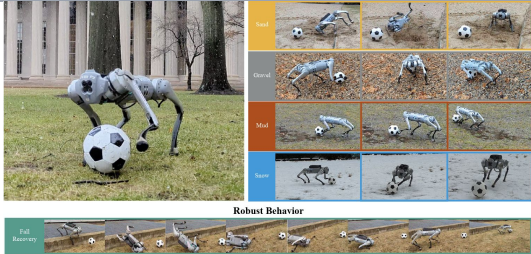
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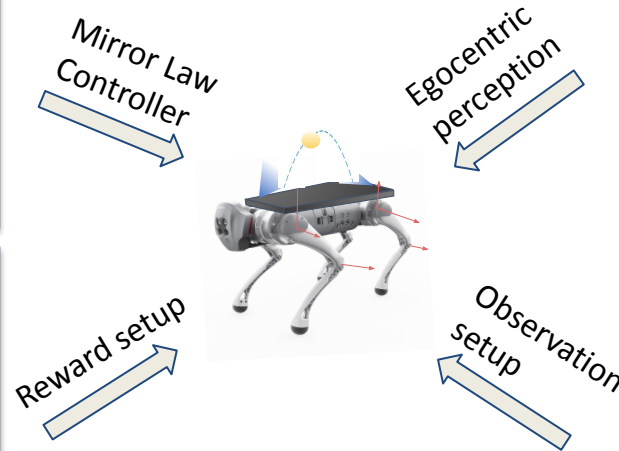
Background / Prior Work



Cassie juggling
Poggensee et al., ECC 2020



DribbleBot
Ji et al., ICRA 2023 — Gaussian centering reward



Legged-manipulator badminton
Ma et al., Science Robotics 2025

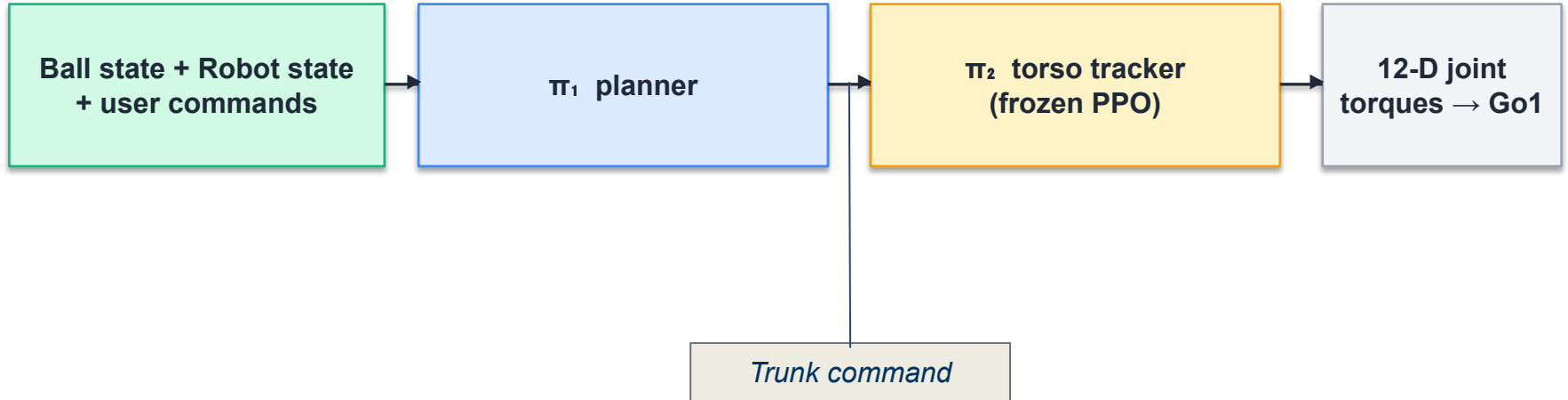


HITTER humanoid table tennis
Su et al., 2025 — hierarchical planning + RL

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Methodology — Hierarchical Architecture



Methodology — Hierarchical Architecture



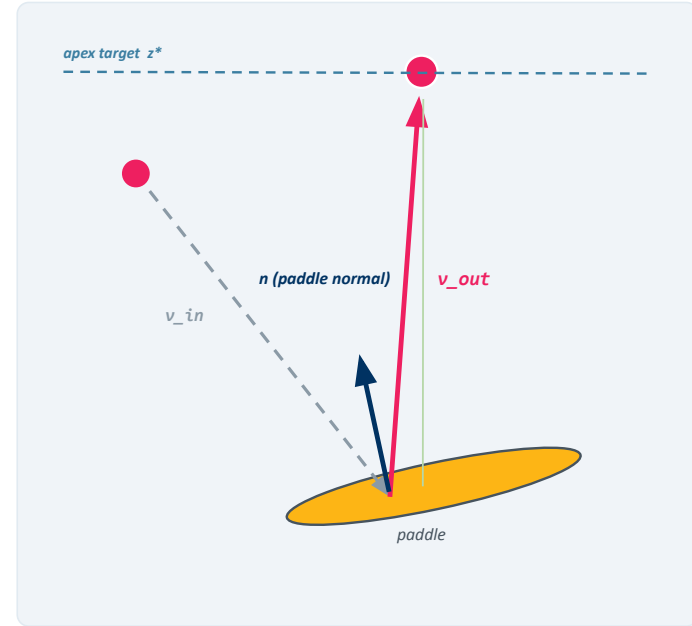
π_1 option A — mirror-law
Analytical controller

π_1 option B — learned planner
PPO policy

Methodology — Mirror-Law Planner (π_1 - Option A)

Key features

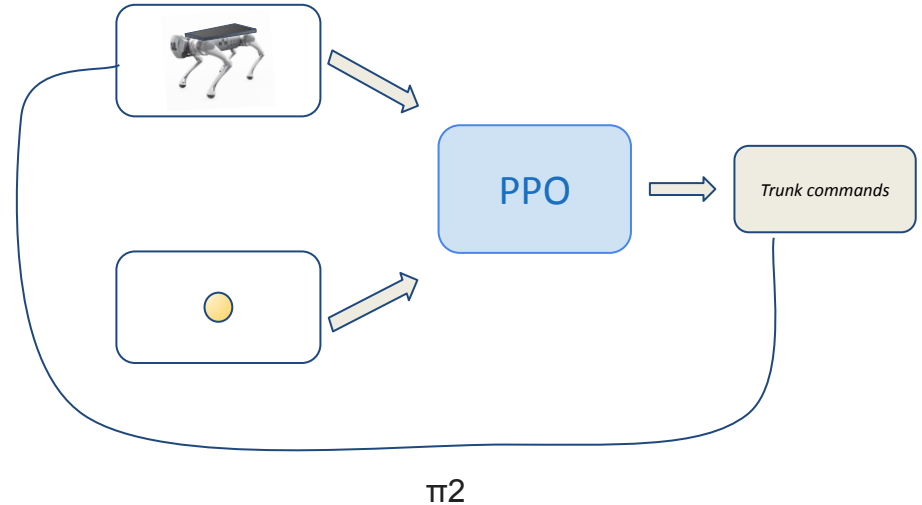
- Mechanical energy adjustment
- Ball state adjustment at apex
- “Reflecting” ball’s velocity components



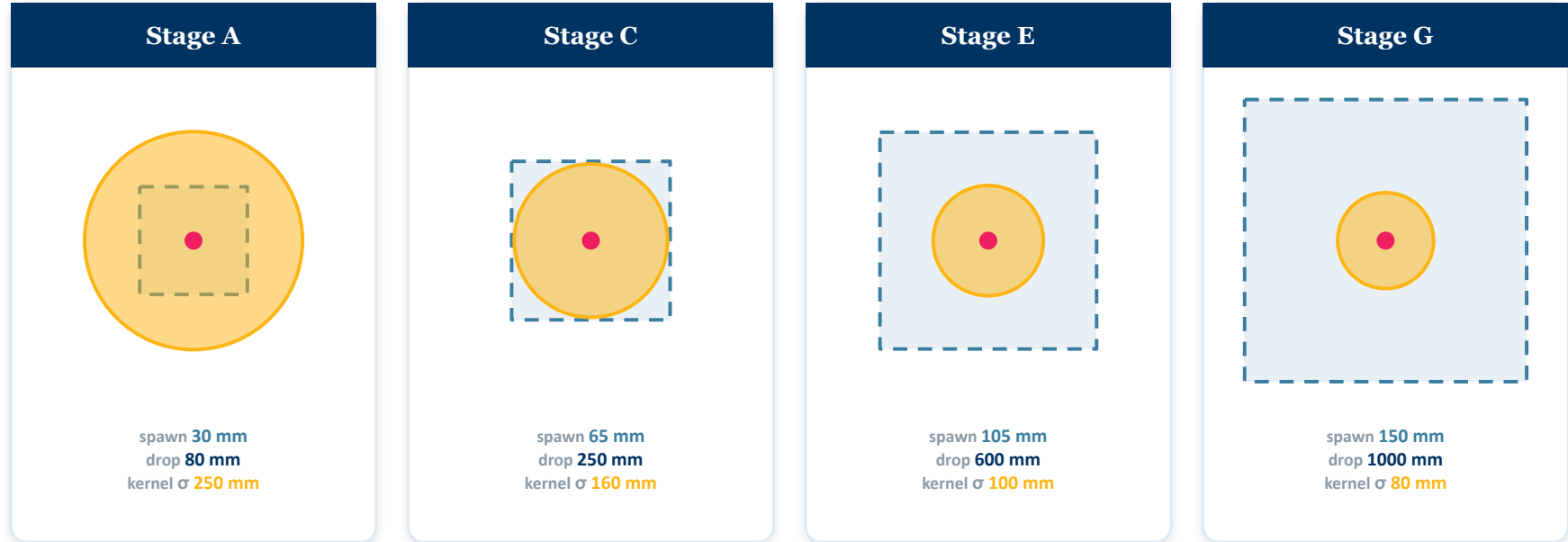
Methodology — PPO Planner ($\pi1$ - Option B)

Key features

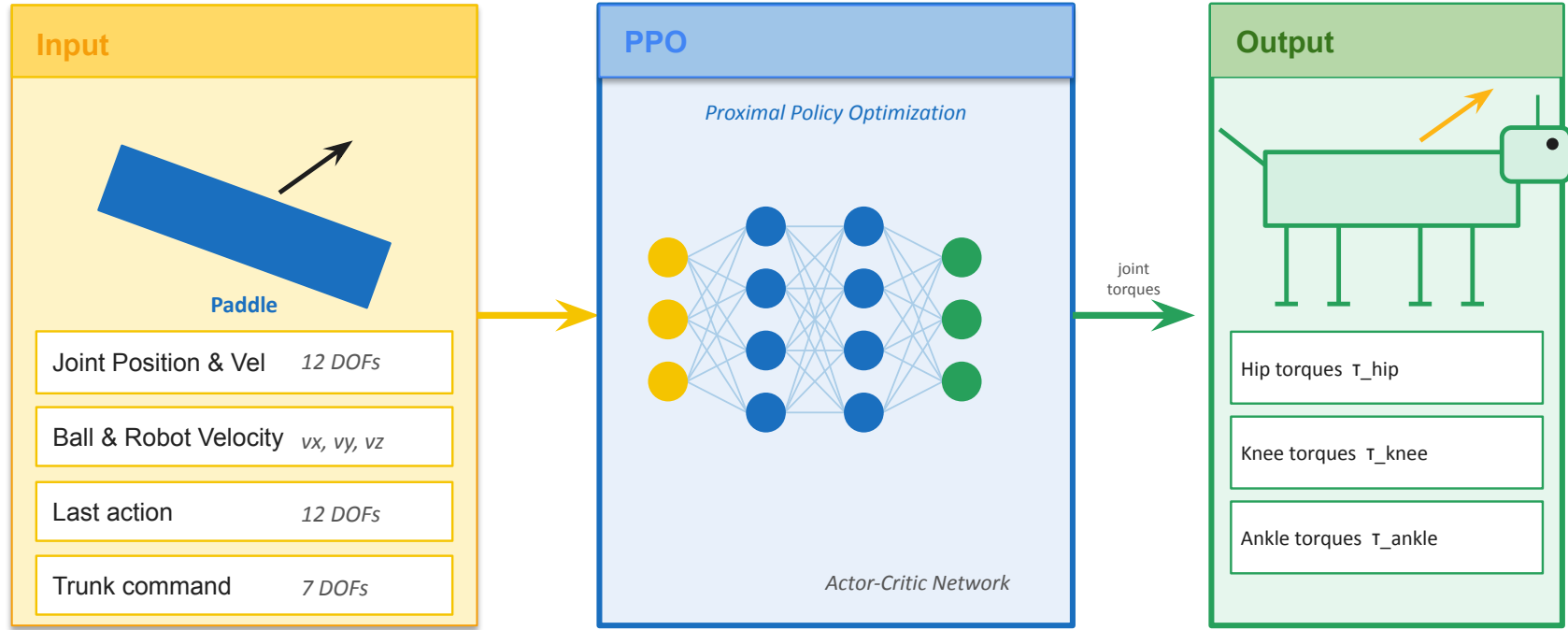
- Robot state input as well as ball input
- Trained with
- Produces end-to-end RL policy



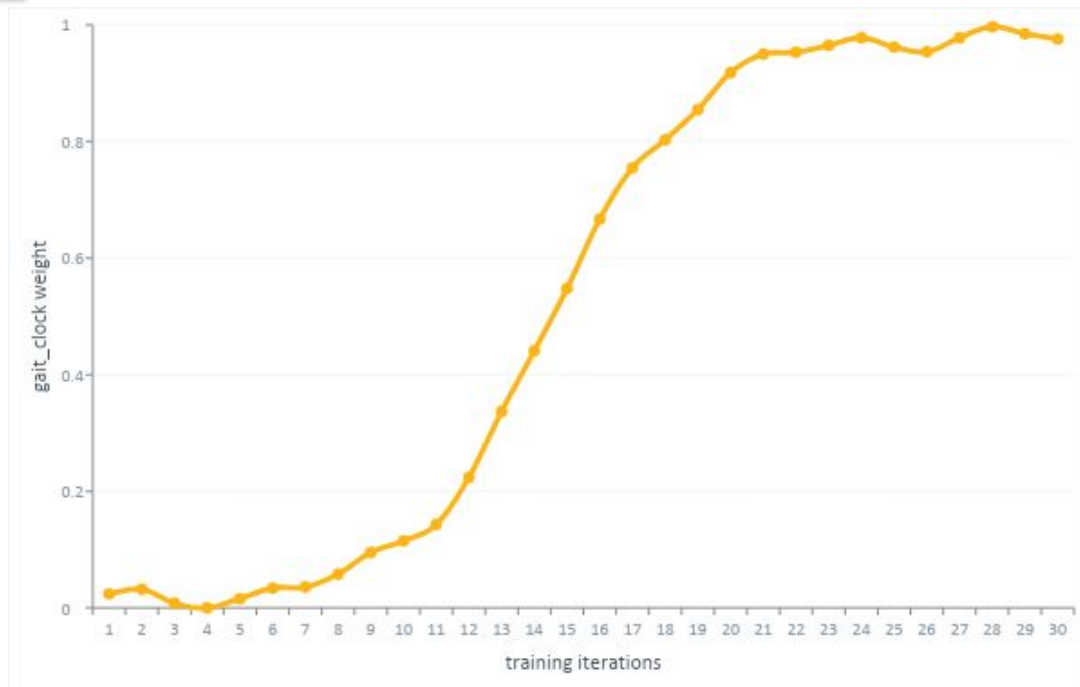
Methodology — Spawning Curriculum



Methodology — Learning-Based Torso Tracker (π_2)



Methodology — Gait Shaping



"Shuffle" failure

Feet never leave the ground.
Micro-adjustments to balance.

Clean 2.5 Hz trot

Policy learns the schedule
rather than micro-shuffling.

WTW reward terms

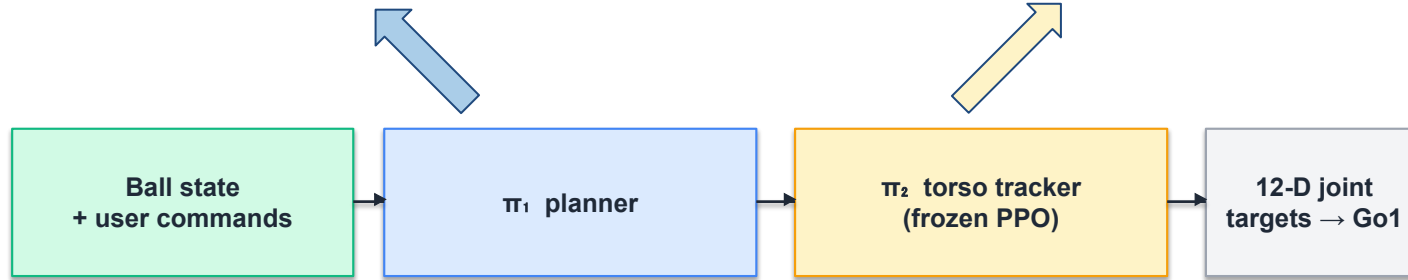
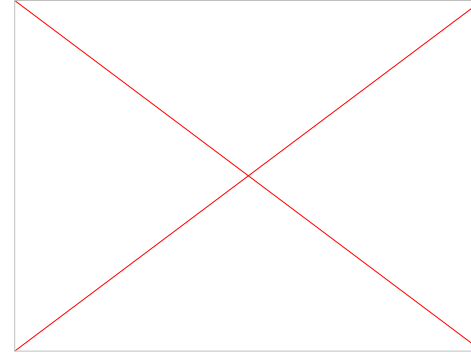
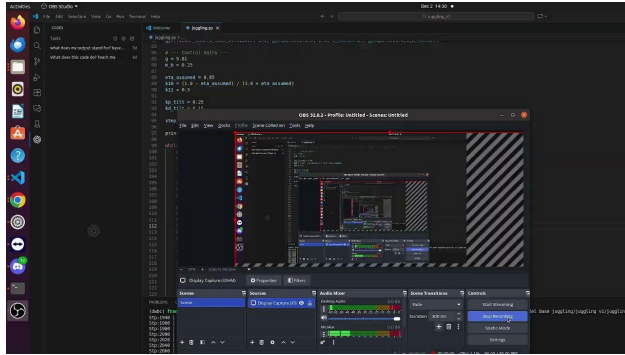
gait_clock	+2.0
feet_air_time	+3.0
foot_clearance	+3.0
feet_slide	-0.5
action_rate_l2	-0.03

Outline

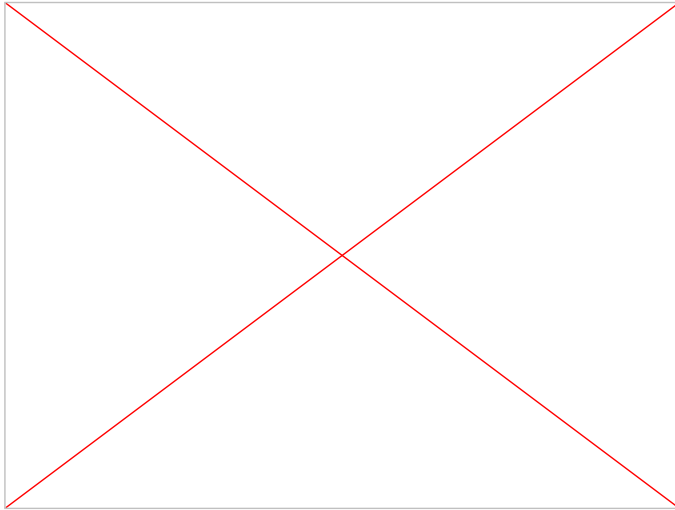
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Mirror Law Results

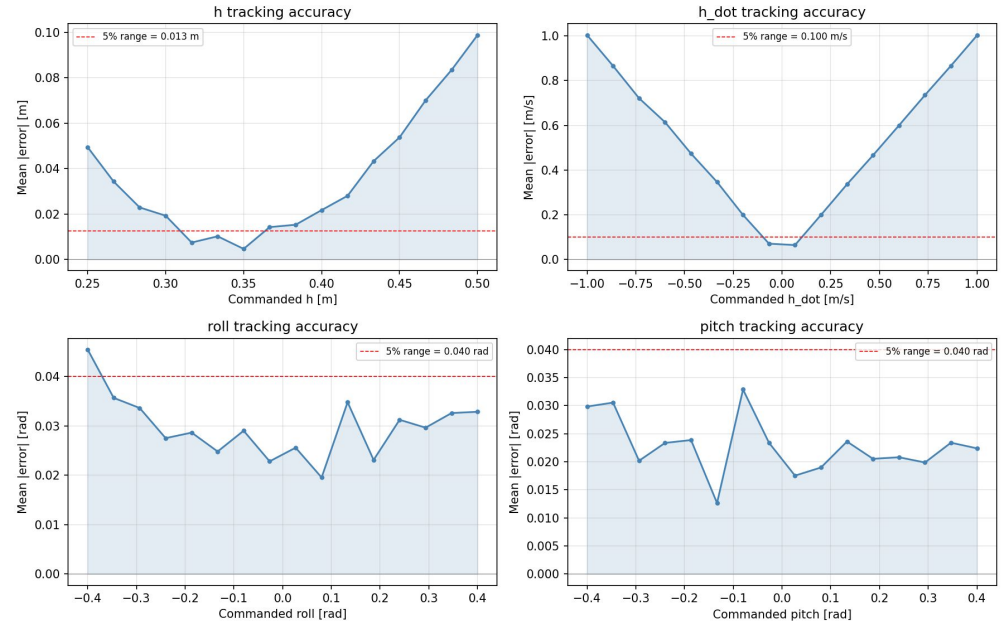
Results — Mirror-law Controller Validation



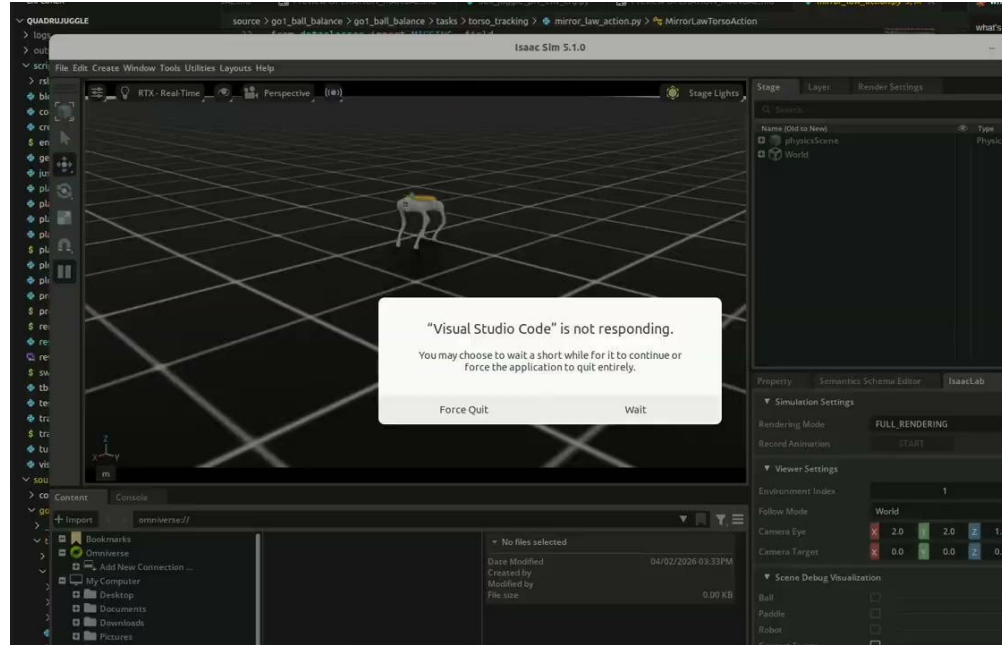
Results — Torso Tracking



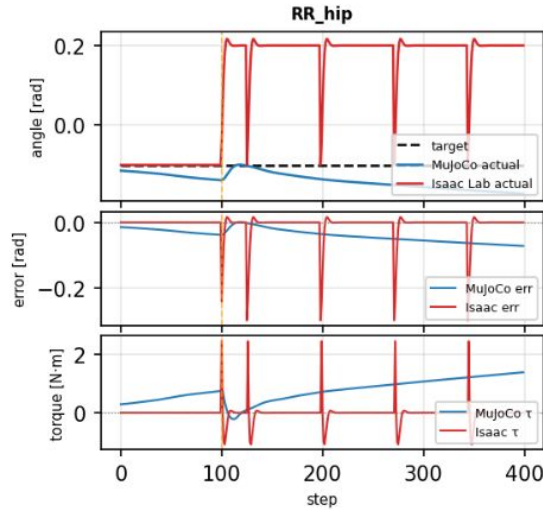
PI2 1D Sweep — Tracking Accuracy per Dimension



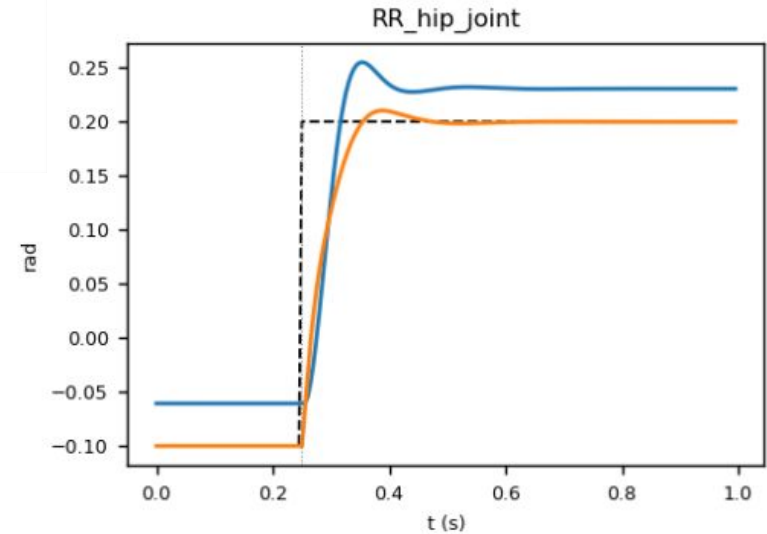
Results — Isaac Lab Simulation



Results — Sim to Sim



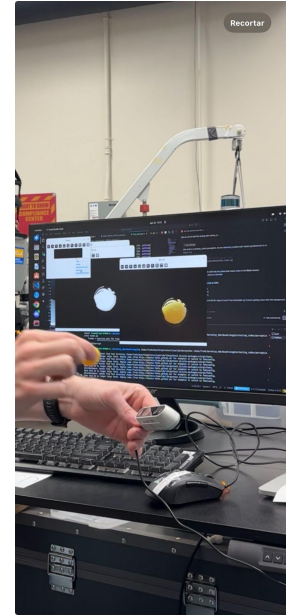
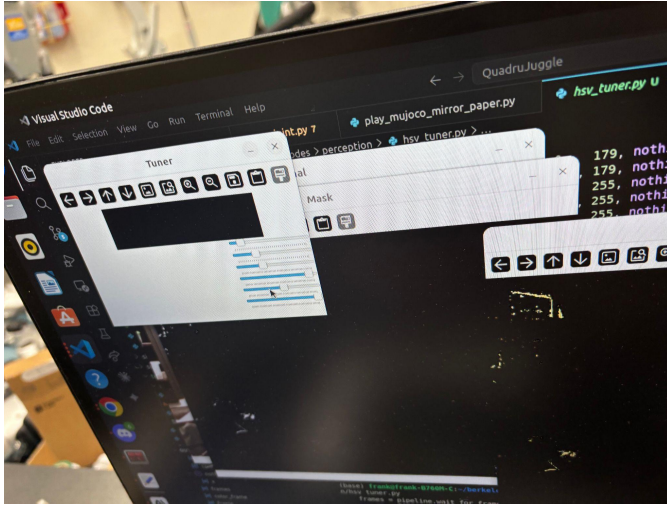
error [rad]



Perception Results

Results — Perception

Color specifics: tuning HSV params



Results — Perception

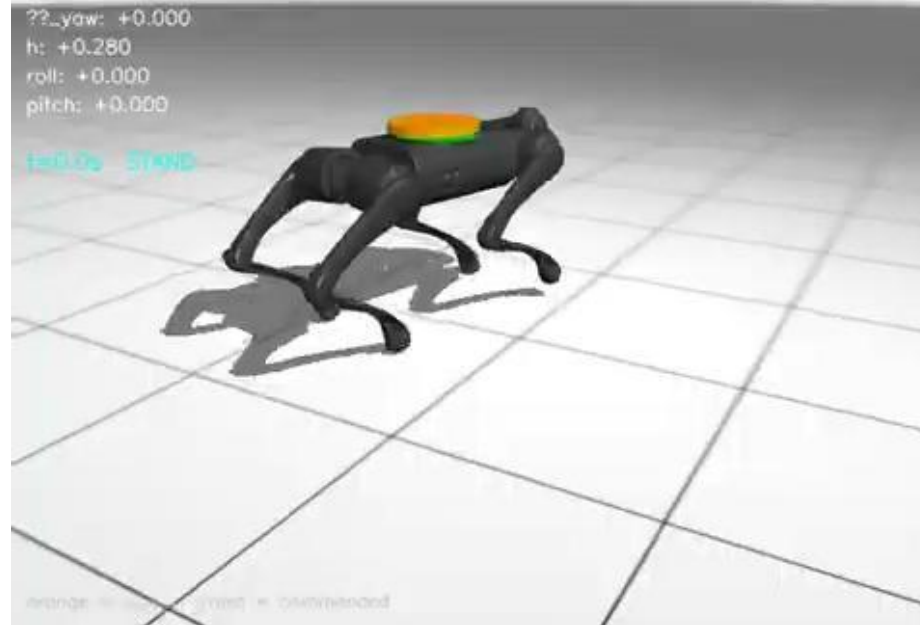
Pipeline

1. Color-based ball detection (OpenCV)
2. Depth sensor: Pixel $\rightarrow$ 3D Position
3. Extended Kalman Filter



End-to-end Learning Results

Results — Velocity/Yaw/Trunk Height + Paddle Orientation



Results — Juggling with Apex Height Variation



Results — Juggling + Balancing Interleaved



Conclusions

- Modular, decoupled planner and tracker
- Mirror-law π_1 : physically interpretable, no training, but much less robust
- Pre-trained RL π_2 : frozen, reusable across tasks
- Static balance + active juggling validated in Isaac Lab + MJLab

